Divorce Prediction and Diabetes Prediction

CSE 4820 - Introduction to Machine Learning
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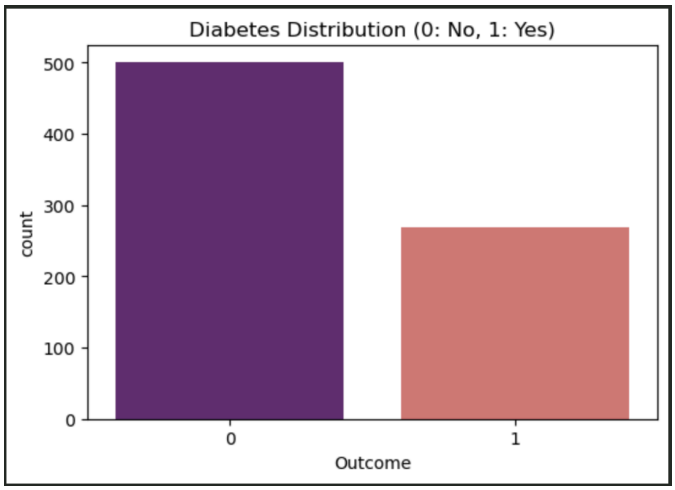
1 Introduction

This report presents a comparative study of two supervised machine learning classification algorithms: Logistic Regression and Random Forest, applied to two distinct datasets: a Diabetes dataset and a Divorce Prediction dataset. The goal is to build predictive models, evaluate their performance, and compare their effectiveness in solving binary classification problems. Both problems are framed as supervised binary classification tasks, where the model learns from labeled training data and predicts one of two possible outcomes for unseen data.

2 Dataset Overview

2.1 Diabetes Dataset

The Diabetes dataset is sourced from the Pima Indians Diabetes Database (available on Kaggle). It contains diagnostic measurements collected from female patients of Pima Indian heritage. The dataset consists of 768 samples and 8 input features, with a binary target variable indicating whether the patient has diabetes.

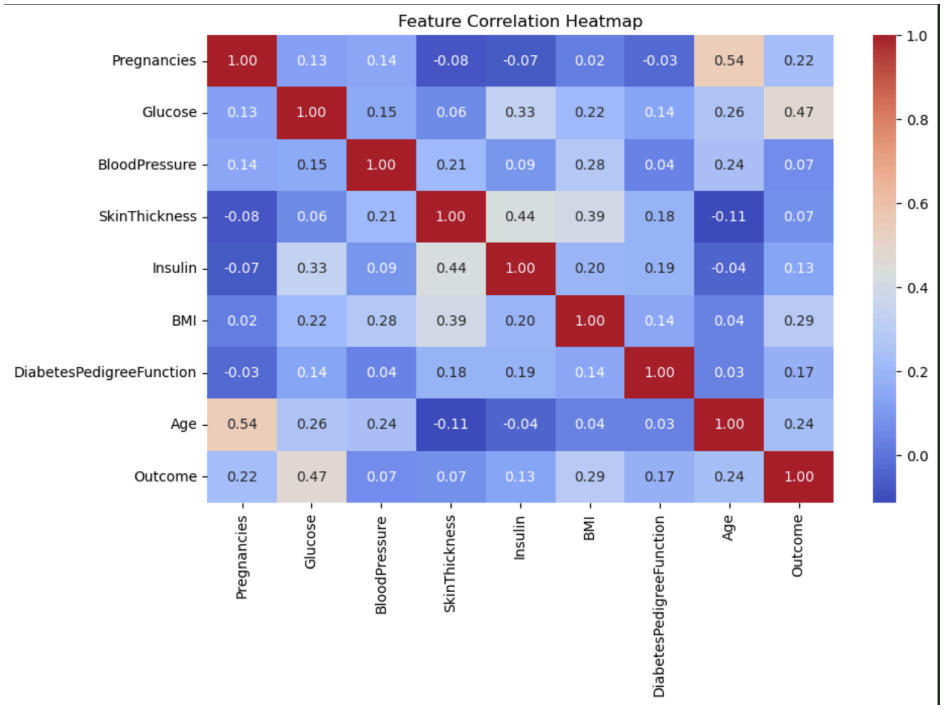


Class 0 (No Diabetes) has 500 samples (65.1%) and Class 1 (Diabetes) has 268 samples (34.9%). The dataset exhibits mild class imbalance, with the negative class (no diabetes) being roughly twice as frequent as the positive class.

Input Features (X):

Feature	Description	Correlation with Outcome
Pregnancies	Number of pregnancies	0.22
Glucose	Plasma glucose concentration	0.49 (highest)
BloodPressure	Diastolic blood pressure (mm Hg)	0.17
SkinThickness	Triceps skinfold thickness (mm)	0.21
Insulin	2-hour serum insulin (mu U/ml)	0.20
BMI	Body mass index	0.31
DiabetesPedigreeFunction	Diabetes pedigree function score	0.17
Age	Age of the patient (years)	0.24

The output variable (y) or the outcome was binary, with 0 being no diabetes and 1 being diabetic.



For the preprossing, zero values in Glucose, BloodPressure, BMI, Insulin, and Skin Thickness were treated as missing data because these values are not physiologically meaningful in a clinical context. These missing values were then replaced using the median of each respective column to reduce the influence of outliers and maintain robustness. After handling missing data, all features were standardized using StandardScaler, transforming them to have zero mean and unit variance to ensure consistent scaling across variables. Finally, the dataset was split into training and testing sets using an 80/20 split, resulting in 614 training samples and 154 test samples, with a fixed random_state=42 to ensure reproducibility of the results.

2.2 Divorce Prediction Dataset

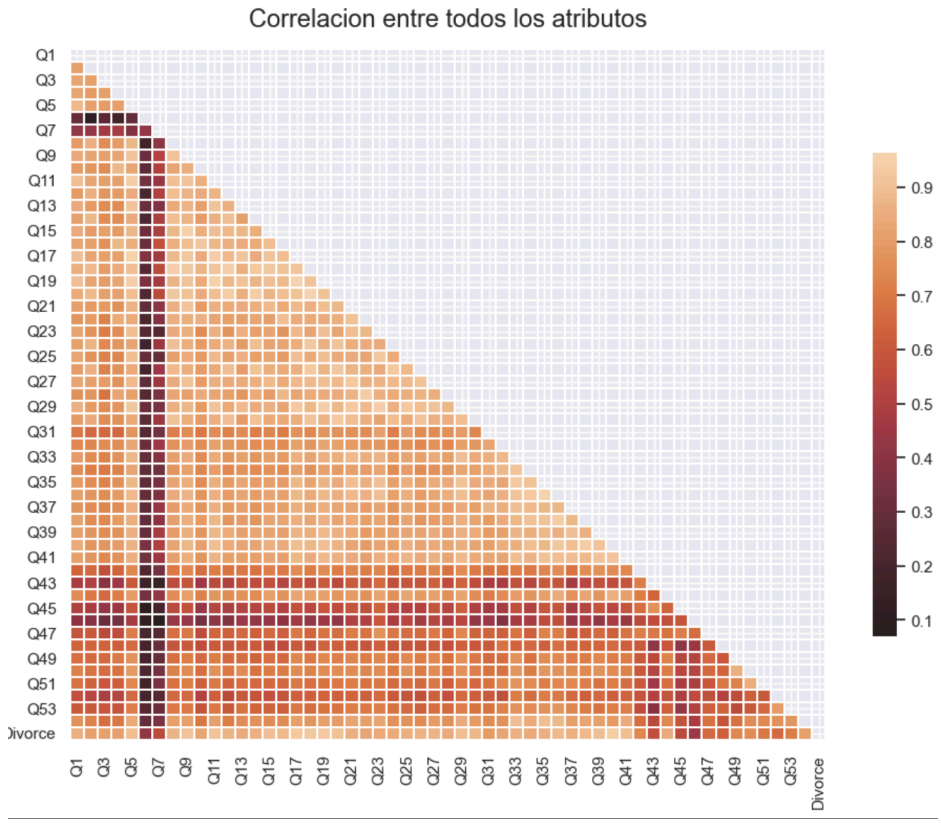
The Divorce Prediction dataset is based on a psychological study using the Divorce Predictors Scale (DPS). It contains responses to 54 questionnaire items (Q1–Q54) answered on a 5-point Likert scale (0–4) by married and divorced couples. The dataset consists of 170 samples.

The dataset shows a nearly balanced class distribution. Class 0 (Not Divorced / Married) contains 86 samples, accounting for 50.6% of the data, while Class 1 (Divorced) contains 84 samples, representing 49.4%. This close balance between the two classes suggests that the dataset does not suffer from significant class imbalance, which helps ensure that model training is not biased toward one class over the other.

Input Features (X): 54 questionnaire features (Q1–Q54) covering five psychological dimensions:

Questions	Psychological Dimension	Example Item
Q1–Q5	Incompatibility & Communication	If one of us apologizes when our discussion deteriorates, the discussion ends.
Q6–Q10	Shared Time & Enjoyment	I enjoy our holidays with my wife.
Q11–Q20	Harmony & Shared Values	My spouse and I have similar ideas about how marriage should be.
Q21–Q30	Knowledge of Partner	I know my spouse's hopes and wishes.
Q31–Q54	Conflict Behavior & Emotional Response	I feel aggressive when I argue with my spouse.

The output variable (y) or the outcome was binary, with 0 being married or not divorced and 1 being divorced.



No missing values were present in the dataset, with all 54 features fully complete across all 170 samples. Since the dataset was already clean, no imputation was required. All features were then standardized using StandardScaler to ensure they were on a consistent scale, improving model performance and stability. Finally, the dataset was split into training and testing sets using an 80/20 ratio, with random_state=0 applied to ensure reproducibility of the results.

3. Problem Definition

Diabetes Problem Statement:

Given a patient's diagnostic measurements (glucose, BMI, age, insulin, etc.), predict whether the patient has diabetes (1) or not (0).

Divorce Problem Statement:

Given a couple's responses to 54 psychological questionnaire items about their relationship behaviors and attitudes, predict whether the couple will divorce (1) or remain married (0).

4. Methodology

4.1 Logistic Regression

Logistic Regression is a linear classification algorithm used for binary outcomes. Instead of directly predicting a class label, it estimates the probability that an input belongs to the positive class using the logistic (sigmoid) function. These probabilities are then converted into class predictions using a decision threshold (commonly 0.5).

Mathematically, it models the log-odds of the target variable as a weighted sum of input features. This makes it interpretable, since each feature has an associated coefficient indicating how strongly it influences the prediction and in what direction (positive or negative effect).

One key assumption is that the relationship between the features and the log-odds of the outcome is approximately linear. Because of this, Logistic Regression tends to perform best when the decision boundary between classes is relatively simple and linear. It is also sensitive to the scale of input features, which is why standardization (using StandardScaler) is important to ensure all features contribute fairly during training.

Compared to more complex models like Random Forest, Logistic Regression is computationally efficient, less prone to overfitting in low-to-moderate dimensional datasets, and easier to interpret. However, it may underperform when the true relationship between features and the target is highly non-linear.

In this implementation, max_iter=1000 was set to ensure the optimization algorithm fully converges, especially for the divorce dataset which has a relatively large number of features (54). Without increasing this limit, the model may fail to converge within the default iterations. The random_state parameter was used to ensure reproducibility, set to 0 for the divorce dataset and 42 for the diabetes dataset so that results remain consistent across runs.

4.1 Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees during training and combines their outputs to make a final prediction through majority voting (for classification). Each individual tree is trained on a bootstrap sample of the dataset (sampling with replacement), and at each split, only a random subset of features is considered. This randomness reduces correlation between trees, which helps lower variance and improves overall generalization performance compared to a single decision tree.

A major advantage of Random Forest is its ability to capture complex non-linear relationships and interactions between features without requiring explicit feature engineering. It is also relatively robust to outliers and noisy data because the ensemble averaging process reduces the influence of any single poor split or tree. In addition, it provides feature importance scores based on how much each feature contributes to reducing impurity across the forest, which makes it useful for interpretability and feature analysis.

Unlike many machine learning models, Random Forest does not require feature scaling, since decision trees are not sensitive to the magnitude of input values. However, scaling was still applied in this project for consistency across models (especially to fairly compare against Logistic Regression).

In this implementation, n_estimators=100 was used, meaning the model consists of 100 decision trees. This provides a balance between performance and computational efficiency. The random_state parameter was set for reproducibility, with 42 used for the diabetes dataset and 0 for the divorce dataset to ensure consistent results across runs.

5. Evaluation Metrics

The following metrics were used to evaluate model performance on the held-out test set:

Metric	Formula	What It Measures
Accuracy	(TP + TN) / Total	Overall proportion of correct predictions
Precision	TP / (TP + FP)	Of all predicted positives, how many were actually positive
Recall (Sensitivity)	TP / (TP + FN)	Of all actual positives, how many were correctly identified
F1-Score	2 × (Precision × Recall) / (Precision + Recall)	Harmonic mean of precision and recall — balances both
Confusion Matrix	2×2 matrix of TN, FP, FN, TP	Detailed breakdown of prediction errors by class

For the diabetes dataset (imbalanced classes), recall for Class 1 (Diabetes) is especially important, as failing to detect a diabetic patient (false negative) has more serious consequences than a false alarm. For the divorce dataset (balanced classes), accuracy and F1-score are equally reliable indicators.

Additionally, the confusion matrices for both datasets are displayed in normalized form (row-wise percentages) alongside raw counts, providing a clearer picture of per-class error rates.

6. Results & Evaluation

6.1 Diabetes Dataset Results

Model	Accuracy	Precision (Class 1)	Recall (Class 1)	F1-Score (Class 1)
Logistic Regression	75.32%	0.67	0.62	0.64
Random Forest	74.03%	0.63	0.65	0.64

Model	True Negative	False Positive	False Negative	True Positive
Logistic Regression	82	17	21	34
Random Forest	78	21	19	36

Several key observations can be made from the results. Logistic Regression achieved slightly higher overall accuracy than Random Forest (75.32% vs 74.03%). However, Random Forest identified more true diabetic cases, giving it higher recall for the positive class (0.65 vs 0.62). Logistic Regression produced fewer false positives, making it slightly more precise. Both models achieved the same F1-score of 0.64 for Class 1, indicating similar overall performance in detecting diabetic patients. The mild class imbalance in the dataset (65% non-diabetic vs 35% diabetic) likely contributed to both models performing better on Class 0 than on Class 1.

6.2 Divorce Dataset Results

Model	True Negative	False Positive	False Negative	True Positive
Logistic Regression	16	0	2	16
Random Forest	16	0	2	16

Model	Accuracy	Precision (Class 1)	Recall (Class 1)	F1-Score (Class 1)
Logistic Regression	94.12%	1.00	0.89	0.94
Random Forest	94.12%	1.00	0.89	0.94

Both models achieved identical performance across all evaluation metrics, with an overall accuracy of 94.12%. Neither model produced any false positives, meaning no married couples

were incorrectly classified as divorced. However, both models missed 2 actual divorce cases, resulting in the same false negatives. The perfect precision score of 1.00 for Class 1 suggests that the Likert-scale questionnaire features are highly effective at distinguishing divorce outcomes. Compared to the diabetes dataset, the much higher accuracy indicates that the divorce dataset contains a cleaner and more directly predictive signal related to the target variable.

7. Feature Importance (Random Forest)

7.1 Diabetes Dataset

Rank	Feature	Importance Score	Notes
1	Glucose	0.2635	Strongest predictor — high glucose is a direct clinical marker of diabetes
2	BMI	0.1641	Body mass index strongly linked to insulin resistance
3	Age	0.1349	Diabetes risk increases with age
4	DiabetesPedigreeFunction	0.1219	Genetic/family history component
5	Insulin	0.0893	Serum insulin levels relate to pancreatic function
6	BloodPressure	0.0840	Hypertension is a comorbidity of diabetes
7	SkinThickness	0.0724	Proxy for body fat distribution
8	Pregnancies	0.0699	Gestational diabetes risk factor

The top three features (Glucose, BMI, Age) together account for 56.25% of the model's predictive power, aligning well with established clinical knowledge about diabetes risk factors.

7.2 Divorce Dataset

Rank	Feature	Importance Score	Description
1	Q11	0.1308	I think one day I'll see that my spouse and I have been in harmony.
2	Q19	0.1052	My spouse and I have similar ideas about roles in marriage.
3	Q18	0.1017	My spouse and I have similar ideas about how marriage should be.
4	Q20	0.0885	My spouse and I have similar values in trust.
5	Q17	0.0866	We share the same views about being happy in our life.
6	Q40	0.0731	We're just starting a discussion before I know what's going on.
7	Q9	0.0601	I enjoy traveling with my wife.
8	Q14	0.0576	Most of our goals for people (children, friends) are the same.
9	Q16	0.0489	We're compatible about what love should be.
10	Q26	0.0274	I know my spouse's basic anxieties.

The top features concentrate heavily in the Harmony & Shared Values dimension (Q11–Q20), suggesting that compatibility in values, life goals, and expectations about marriage are the strongest predictors of divorce. This is consistent with psychological research, which identifies value misalignment as a key driver of marital breakdown.

Notably, Q40 (conflicts starting suddenly) from the Conflict Behavior dimension also ranks highly, indicating that impulsive or poorly managed conflict is another strong divorce predictor.

A large number of features (Q22–Q24, Q35, Q37, Q42, Q45–Q46, Q48–Q54) received near-zero importance scores, suggesting significant feature redundancy in the 54-question survey.

8. Model Comparison

Criterion	Logistic Regression	Random Forest
Model Type	Linear, parametric	Non-linear, ensemble (100 trees)
Diabetes Accuracy	75.32%	74.03%
Divorce Accuracy	94.12%	94.12%
Interpretability	High — coefficients are directly interpretable	Medium — requires feature importance plots
Feature Scaling Required	Yes	No (applied for consistency)
Handles Non-linearity	No	Yes
Overfitting Risk	Low	Low (due to bagging)
Training Speed	Fast	Slower (100 trees)
Feature Importance	Via coefficients only	Built-in importance scores

On the diabetes dataset, Logistic Regression marginally outperforms Random Forest in accuracy (75.32% vs 74.03%), which is somewhat surprising given that Random Forest is generally more powerful. This may be because: (1) the dataset is relatively small (768 samples), giving Random Forest less data to learn complex patterns; and (2) the class imbalance slightly disadvantages tree-based methods.

On the divorce dataset, both models achieve identical results. This is likely because the Likert-scale features provide a very strong, near-linear signal, couples who answer very differently from their partner on relationship satisfaction items are clearly separable, making both linear and non-linear models equally effective.

The Random Forest's feature importance analysis adds significant interpretive value in both cases, revealing that Glucose dominates diabetes prediction and that shared marital values dominate divorce prediction, which Logistic Regression alone cannot provide as clearly.

9. Conclusion

This report demonstrated the use of Logistic Regression and Random Forest for two binary classification tasks: diabetes diagnosis and divorce prediction. Overall, both models performed well, though their effectiveness depended on the characteristics of each dataset.

For the diabetes dataset, Logistic Regression achieved slightly higher accuracy than Random Forest (75.32% vs 74.03%), while Random Forest showed stronger recall for identifying diabetic patients. In the divorce dataset, both models achieved the same high accuracy of 94.12%, suggesting that the survey-based relationship features provide a very strong predictive signal.

Feature importance analysis from the Random Forest model showed that Glucose, BMI, and Age were the most influential predictors for diabetes, while questions related to shared values and marital expectations (Q11, Q17–Q20) were among the strongest predictors for divorce outcomes.

The divorce dataset produced stronger results because its features were more directly related to the target variable, while the diabetes dataset was slightly more challenging due to mild class imbalance. Future improvements could include using techniques such as SMOTE or class weighting to better handle imbalance, applying cross-validation for more reliable evaluation, and experimenting with additional models such as Support Vector Machines or Gradient Boosting.

10. Reference

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